FINAL REPORT

Executive Summary:

This report presents the findings and recommendations resulting from an optimization project conducted to enhance the efficiency of delivery routes for a logistics company. The objective was to identify the optimal delivery routes that minimize both travel time and delay likelihood, considering varying levels of resource allocation. Through mathematical modeling and computational analysis, we explored the impact of different parameters on route optimization and provided actionable insights for operational decision-making.

Project Details:

The optimization problem addressed in this project pertains to identifying the shortest time delivery routes for a logistics company. The goal was to minimize both travel time and delay likelihood while optimizing resource allocation. The project involved reviewing the problem statement, analyzing the provided data, formulating a mathematical model, implementing the model using appropriate software tools, and composing a comprehensive report summarizing the findings and recommendations.

Methodology:

The optimization problem was approached using a combination of mathematical modeling techniques using binary variables and computational analysis. We employed mathematical optimization methods (mainly Integer Programming) to formulate the problem as a mathematical model, incorporating constraints and objectives to represent the delivery route optimization problem. The model was then solved using appropriate software tools (Simplex method) to identify the optimal solution.

Results and Analysis:

The analysis revealed that the optimal delivery routes are influenced by various factors, including distance between cities, delay likelihood (further influenced by parameter beta). By systematically exploring different values of beta, we observed a transition in the optimal routes, highlighting the trade-off between travel distance and delay likelihood. Specifically, as beta increased, the optimal routes shifted from shorter distances with higher delay likelihood to longer distances with lower delay likelihood.

Conclusion:

In conclusion, our experiments reveal that in real-life scenarios, the distances between cities remain constant, indicating that alfa is always equal to 1. By running a loop for beta within the range of [0, 1], we observe a transition in the optimal path from "A → D → B → O" to "A → D → E → O" as beta increases from 0.1 to 0.2. To gain deeper insights into the range of beta values where the optimal path changes, we conduct a secondary loop within the range of [0.1, 0.2]. Our research indicates that this transition occurs between beta = 0.16 and beta = 0.17. At beta = 0.16, the optimal path is "A → D → B → O" with a total distance of 75.0 miles and a delay likelihood of 0.008937. However, at beta = 0.17, the optimal path shifts to "A → D → E → O" with a total distance of 96.0 miles and a delay likelihood of 0.002703.

Ultimately, the choice between these two options rests on the company's resources. Opting for a shorter travel distance of 75 miles may result in a higher delay likelihood of 0.008937, whereas selecting the longer route of 96 miles offers a lower delay likelihood of 0.002703. The decision should

be made considering the company's financial resources and the trade-off between travel distance and delay likelihood.

Recommendations:

Based on the findings of the optimization analysis, we recommend that the logistics company consider the trade-off between travel distance and delay likelihood when determining optimal delivery routes. By strategically adjusting resource allocation (beta parameter), the company can optimize route efficiency while minimizing delay likelihood, thus enhancing overall operational performance.

Acknowledgments:

We would like to express our gratitude to Amazon for providing the opportunity to work on this optimization project and for their collaboration throughout the process. We also extend our appreciation to Professor Tom Asaki for guidance and feedback during the project.

References:

Efficient Shortest Path Simplex Algorithms on JSTOR